



MMA093 Rigid Body Dynamics

Assignment 3

Supplementary Part

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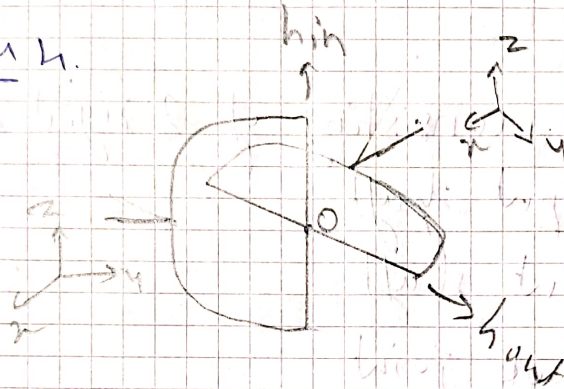
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1 Problem 4: Derive the Analytical relation

Problem 4.



Let o be the center of hook.

Input shaft axis is z axis $Z_{in} (0, 0, 1)$

Output shaft axis is tilted by angle β about y -axis

$$Z_{out} = (\sin \beta, 0, \cos \beta).$$

Input fork initial vector along x . Rotating around input z -axis $u(1, 0, 0)$

$$h_{in}(\theta) = \begin{pmatrix} \cos \theta \\ \sin \theta \\ 0 \end{pmatrix} \quad \text{--- (1)}$$

Output fork initial vector along y . Using Rodriguez's rotation formula $N = (0, 1, 0)$

$$h_{out}(\phi) = N \cos \phi + (Z_{out} \times N) \sin \phi$$

$$\begin{aligned} \mathbf{z}_{out} \times \mathbf{v} &= (\sin \beta, 0, \cos \beta) \times (0, 1, 0) \\ &= (-\cos \beta, 0, \sin \beta). \end{aligned}$$

$$\mathbf{h}_{out}(\phi) = \begin{pmatrix} -\cos \beta \sin \phi \\ \cos \phi \\ \sin \beta \sin \phi \end{pmatrix} \quad - (2)$$

Hook condition:

Two fork orientation to be orthogonal

$$\mathbf{h}_{in} \cdot \mathbf{h}_{out} = 0.$$

$$\mathbf{h}_{in} \cdot \mathbf{h}_{out} = (\cos \theta, \sin \theta, 0) \cdot (-\cos \beta \sin \phi, \cos \phi, \sin \beta \sin \phi).$$

$$= -\cos \beta \cos \theta \sin \phi + \sin \theta \cos \phi + 0.$$

$$= -\cos \beta \cos \theta \sin \phi + \sin \theta \cos \phi = 0$$

$$\sin \theta \cos \phi = \cos \beta \cos \theta \sin \phi \quad - (3)$$

Divide (3) by $\cos \theta \cos \phi$ both sides.

$$\tan \theta = \cos \beta \tan \phi.$$

position angle relation b/w input angle θ & output angle ϕ is

$$\tan \phi = \frac{1}{\cos \beta} \tan \theta \Rightarrow \tan \theta = \cos \beta \tan \phi$$

differentiating

$$\phi(\theta) = \tan^{-1}(\cos \beta \tan \theta) \quad - (4)$$

Differentiate ④ with respect to t

$$\frac{d\phi}{dt} = \frac{1}{1 + (\cos\beta \tan\theta)^2} \cdot \cos\beta \frac{d}{dt}(\tan\theta)$$

$$\frac{d}{dt} \tan\theta = \sec^2\theta \dot{\theta} = (1 + \tan^2\theta) \dot{\theta}$$

$$\dot{\phi} = \frac{\cos\beta (1 + \tan^2\theta) \dot{\theta}}{1 + \cos^2\beta \tan^2\theta}$$

$$\dot{\theta} = \omega_{in} \quad \dot{\phi} = \omega_{out}$$

$$\boxed{\frac{\omega_{out}}{\omega_{in}} = \frac{\cos\beta (1 + \tan^2\theta)}{1 + \cos^2\beta \tan^2\theta}}$$